

1 Importance of HVAC Noise in Wards

In a clinical ward environment, the tolerance for background noise is exceptionally low. Moving from a baseline NC 30 up to an NC 45 range represents a critical transition from a "restful healing environment" to a "communication hazard," where mechanical noise actively masks direct speech and critical medical alarms.

When simulating an elevated HVAC noise floor, you are modeling the real-world impact of mechanical intrusion on patient recovery—specifically how increased system stress directly degrades the Signal-to-Noise Ratio (SNR) and speech intelligibility at the patient's bedside.

NC Ratings: Clinical Healthcare Environments

NC Rating	Experience in a Gym	Simulation Context
NC-30	Ideal. Imperceptible noise. Clear communication.	"Optimal" state for patient recovery.
NC-35	Noticeable. Partial masking. Elevated voices required.	"Stressed" standard with acoustic interference.
NC-45	Unsafe. Severe masking. Disrupts medical alarms.	"System Failure" causing near-zero intelligibility.

Main factors affecting NC:



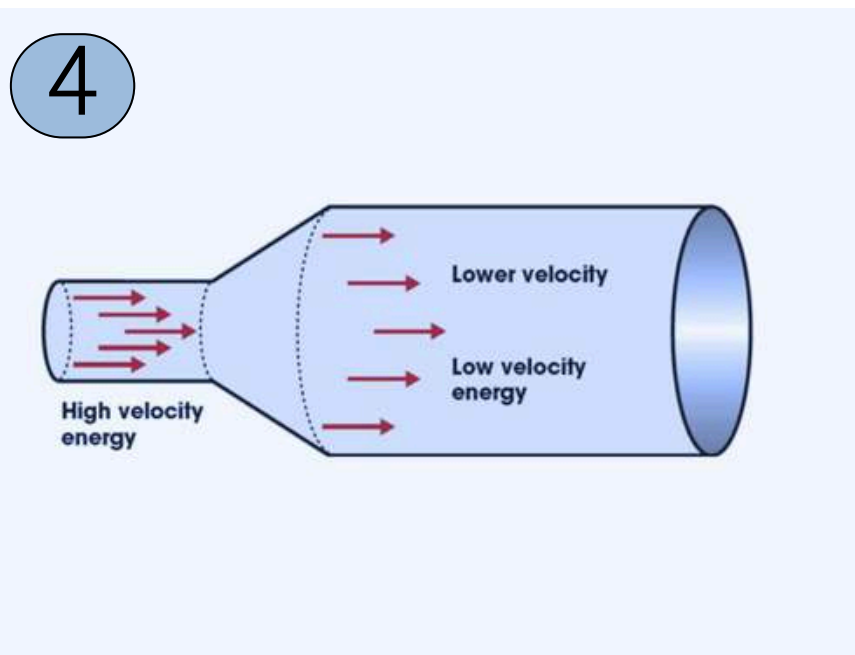
HVAC diffusers and air outlets



Ducts, fans, and mechanical equipment



External noise through windows and facade openings



Airflow velocity and duct friction

Acoustic Evaluation Parameters

Early Decay Time (EDT):

Derived from the first 10 dB of sound decay, EDT reflects the perceived reverberation condition in the room.

Speech Transmission Index (STI):

STI is an objective measure of speech intelligibility, showing how well speech is transmitted across the selected frequency bands.

Clarity (C50):

C50 compares early sound energy (0-50 ms) with late reverberant energy, indicating how clearly speech, especially consonants, can be understood.

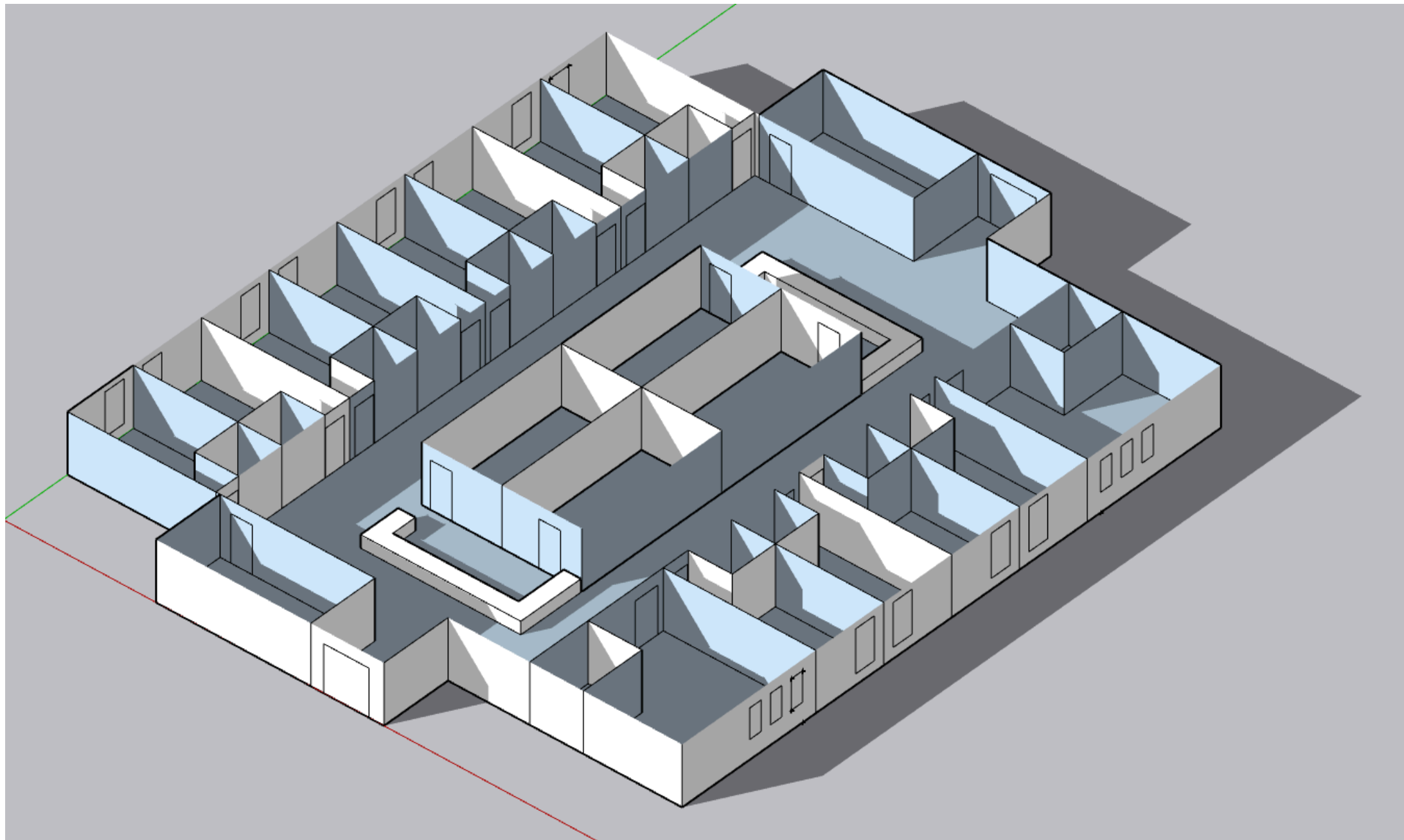
Reverberation Time (T30):

Calculated over a 30 dB decay range, T30 represents the objective persistence of sound, particularly within the 500-2000 Hz speech range.

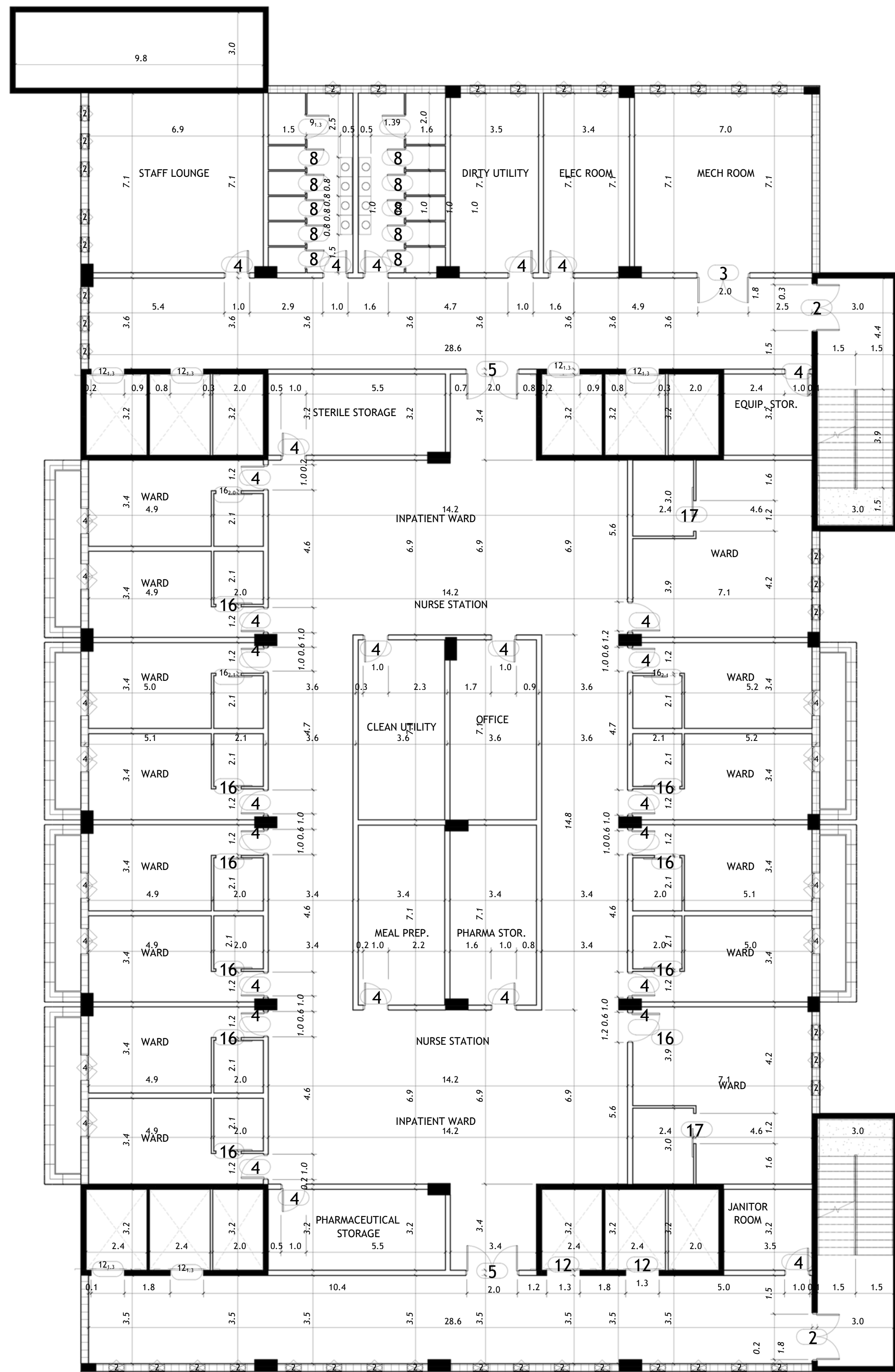
2 Patient Wards Plan & Section & 3d VIEW

Architectural Configuration and Acoustic Vulnerability

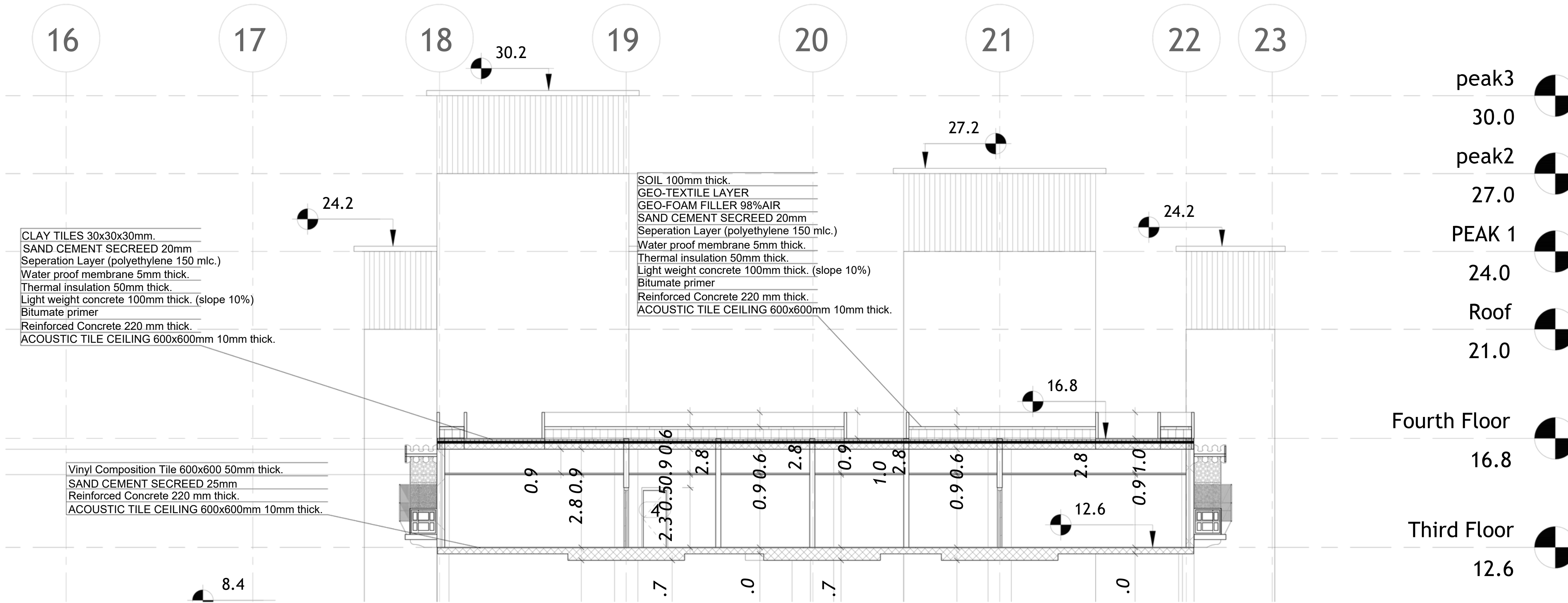
The architectural configuration of the ward heavily dictates its baseline acoustic performance. The floor plan reveals a centralized, high-traffic clinical core—comprising utility zones, meal preparation, and dual nurse stations—surrounded by peripheral patient recovery rooms. This layout inherently risks acoustic coupling, where noise from active staff hubs can propagate directly into quiet patient zones. Vertically, the section details a standard clinical clear height of 2.8 meters, bounded by highly reflective surfaces including 50mm vinyl composition floor tiles over 220mm reinforced concrete slabs, and solid wall partitions. Although capped with a 10mm acoustic tile ceiling, this specific geometric and material assembly lacks sufficient high-frequency absorption. This rigid architectural shell is the primary driver of the persistent 2000 Hz reverberation "bump" observed in the EDT and T30 analyses, proving that the physical room intrinsically traps acoustic energy even before mechanical HVAC noise is introduced.



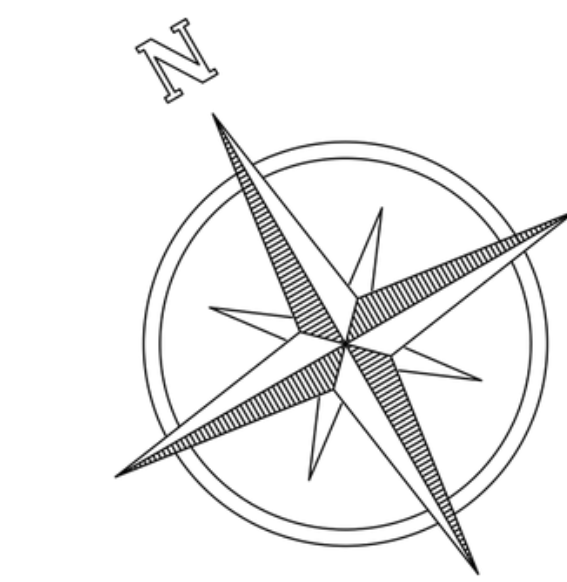
3D VIEW



Wards Plan



Wards Section



ALL UNITS IN METER



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INSTRUCTORS:

Prof. Othman Alshamrani (Co)
Prof. Mohammed Essam Shaawat
Dr. M. Abdul Mujeebu
Dr. Mohammad Farooqi
Dr. Saiful Islam
Dr. Abdulaziz Almohassen
Dr. Mohammed Alkhater
Dr. Musaad Al-hawas
Dr. Fahad Alyami
Dr. Hany Hassam Eldien
Dr. Abdullah Qahtani
Dr. Mohammed Alkhater
Engr. Noman Ashraf
Engr. Meqdad Hasan
Engr. Rehan Jamil

DONE BY:

ABDULELAH ALQABBANI

PROJECT:

Emergency Hospital

DRAWING NAME:

ACOUSTIC SIMULATIONS

SCALE:

1:150

SHEET NUMBER:

A65